PROBLEM STATEMENT, LITERATURE REVIEW & COMPETITOR ANALYSIS

MScFE Capstone Project

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1. Problem Statement:

1.1. Background

Credit-risk modeling goes beyond just distinguishing “good” vs. “bad” borrowers. A good credit risk model needs to (a) have good rankings, (b) give useful probabilities for decision making, and (c) explain itself in a way understandable by risk managers, auditors, and affected applicants. In the context of regulatory model-risk management, validation includes assessing conceptual soundness, outcomes, benchmarking, documentation, and monitoring; the new 2026 Federal Reserve guidelines replace SR 11-7 while retaining all essential requirements.

According to the literature, machine learning models are known to increase the performance of consumer credit risk models due to their ability to capture the nonlinearities and interactions that could escape detection using more simplistic parametric models (Khandani et al., 2010; Lessmann et al., 2015). However, predictive performance cannot be the only criterion for evaluating credit risk models. A model might demonstrate high discrimination but at the same time have poor calibration, use information unavailable at the time of decision, or generate explanations that are irrelevant to the factors involved in the actual lending decision.

1.2. Problem definition

Initially, the project used the OpenIntro Lending Club dataset snapshot, containing about 10,000 observations. While this set was sufficient to illustrate binary classification, it was not enough for conducting a proper credit risk study because of the multiple loans issued recently with unknown `current` statuses. Considering such loans as non-defaults leads to immature or ambiguous target variable.

In the updated project, the subset of accepted-loans from the comprehensive historical Lending Club data is utilized. The original source consists of 2,260,701 observations of accepted loans and 151 variables describing the loans issued between 2007 and 2018 Q4. To get a clearer picture of the target variable, 36-month loans issued during 2011-2015 with statuses "`Fully Paid`" or "`Charged Off`" are currently analyzed. The resulting prepared modelling cohort contains 603,589 loans: 519,362 fully paid loans and 84,227 charged-off loans.

The target is defined as:

$$Y = \begin{cases} 1, & \text{if loan status is Charged Off,} \\ 0, & \text{if loan status is Fully Paid.} \end{cases}$$

This is the observed final loan outcome, not a one year default probability. Statuses like `Current`, `Late` and `In Grace` Period are not included as a non-default status because including them increases outcome ambiguity and the study does not model censored data or do reject inference.

1.3. Research problem

The central problem is:

> How can an explainable credit-risk modelling framework deliver out-of-time default predictions from historical Lending Club application and credit history data without causing target leakage, considering temporal validation, distinguishing between discrimination and probability calibration, and separating borrower level risk modelling from the economic decision problem?

The solution to this problem will include:

- Historically resolved 36-month loan cohort.
- Chronological training, validation, and test design.
- Separation of application/credit variables and repayment outcomes.
- Baseline features excluding Lending Club's own grade and pricing variables.
- Comparison features including `grade`, `sub_grade`, `int_rate`, and `installment`.
- Training-only imputation, scaling, and categorical encoding.
- Evaluation based on discrimination and calibration metrics instead of ROC-AUC only.
- Explainability of the final chosen model.
- Fairness and Expected Loss extension discussion constrained to its limits.

1.4. Research objectives

The project will be completed in accordance with five goals:

1. Develop model of Probability of Default with controlled target leakage using application and credit history variables.
2. Comparison of an explainable baseline with nonlinear ensemble models.
3. Validation of the model based on out-of-time evaluation instead of just train-test random split.

4. Evaluation of the probability calibration of the model with Brier Score and calibration curves.
5. Analysis of the model explainability and operational decision rules' potential.

Conceptually Expected Loss can be expressed as:

$$EL = PD \times LGD \times EAD.$$

Nevertheless, the currently developed data preparation stage cannot yet be regarded as an LGD and EAD model because these parts should be provided as an intended extension unless other notebooks develop and validate them.

1.5. Research questions

1. How do Logistic Regression and nonlinear ensemble models compare for predicting observed charge-off outcomes?
2. How do the results for out-of-time performance differ from those for validation performance depending on loan vintage?
3. How do the borrower/application baseline and a feature set including the Lending Club's own grade and pricing differ in their performance?
4. Are the obtained probabilities properly calibrated to be used in further risk assessment?
5. What effect do the choices of class weighting and resampling have on discrimination, calibration, and minority-class identification?
6. Is it possible to find interpretable and stable predictors for the given model?
7. How can a PD model be linked to expected loss calculation without using LGD and EAD estimations?

1.6. Data source and cohort

In this research, the accepted-loans file, `accepted_2007_to_2018Q4.csv.gz`, from the All Lending Club loan data set created by Nathan George on Kaggle (George, 2019), was used. We chose 603,589 loans fully paid back or charged off during 36 months issued in 2011-2015: 519,362 loans were marked as `Fully Paid` and 84,227 as `Charged Off`. Loans with other statuses were excluded from the modelling cohort rather than classified as non-default. The outcome variable is observed loan status, but not default within 12 months period.

2. Literature review

2.1. Credit scoring and interpretable baselines

Statistical scorecards and Logistic Regression have been used extensively in credit scoring since they allow an interpretable approach with relatively stable implementation and familiar governance. In particular, Logistic Regression serves as a valuable benchmark due to the interpretability of the coefficients that reflect directional feature effects and allow the comparison of complex algorithms to this simple reference point.

The classical research in credit-risk area also established the importance of financial and borrower level features in predicting the quality of repayment. Altman's discriminant analysis proved the value of structured financial indicators in predicting defaults (Altman, 1968). Similarly, Khandani et al. (2010) used nonlinear machine learning algorithms for improving consumer-credit forecasts.

However, in this project Logistic Regression shouldn't be abandoned but should remain a transparent benchmark to evaluate complex models.

2.2. Ensemble machine learning

The Ensemble Methods like Random Forest or Gradient Boosting Machines (GBMs) are able to capture nonlinear relationships and interactions without requiring all relationships to be coded manually. The benchmarking performed by Lessmann et al. (2015) of a wide range of classification methods on credit-scoring data revealed the strong performance of ensemble techniques relative to the traditional methods and underlined the necessity to evaluate predictions based on their managerial value instead of statistical accuracy only.

The algorithm is most pertinent to the current case as it is based on gradient-boosted decision trees and aimed at computational efficiency (Ke et al., 2017). At the same time, high performance does not make an ensemble a good choice for a credit model; it must be evaluated in terms of calibration, leakage, stability, explainability, and appropriateness.

Thus, the current project takes the approach of considering Logistic Regression as the benchmark and evaluating one or more nonlinear models as possible extensions. "Using XGBoost or LightGBM" would not be a research contribution itself as such combinations are widely known. The contribution of the project is rather methodological comparison within explicitly chronological and leakage-free evaluation process.

2.3. Data leakage and temporal validation

There are inherent features in credit outcomes that have a temporal component: borrower pools, underwriting criteria, interest rates, economic conditions, and loan products can change. Random split can divide similar vintages between training and test partitions and make evaluation unrepresentative of deployment in the future.

The current project addresses this problem by using issue month as the basis for:

- Training: 2011–2013.
- Validation: 2014.
- Held-out test: 2015.

The observed charge-off rates are 12.52% in training, 13.73% in validation, and 14.89% in the held-out test period. The temporal shift illustrates the need for time-related performance analysis during validation.

Also, the current project distinguishes origination information and post-origination repayment variables. Variables like `total_rec_prncp`, `recoveries`, and `collection_recovery_fee` are the indicators of repayment or recovery outcomes and kept separately in the loss-outcome table instead of using them as predictors for PD. This is a key step as leakage happens when information unavailable at prediction time enters model development process.

2.4. Class imbalance and calibration

The prepared cohort is biased in that it includes 84,227 charged-off loans and 519,362 fully paid loans. Therefore, accuracy is insufficient as a measure since a model can predict nearly all loans to be non-default yet be considered accurate without actually identifying risky loans.

Possible solutions:

- Class-weighted learning.
- Training-only resampling.
- Threshold adjustment.
- Evaluation with precision, recall, PR-AUC, ROC-AUC, and confusion matrices.

However, risk assessment involves more than ranking of cases. PD values are treated as probabilities; hence, calibration is essential. A model could rank risky loans efficiently but would consistently overestimate or underestimate the likelihood of an event. Calibration, thus, should be

assessed through Brier score, reliability diagrams, calibration slope/intercept if possible, and results obtained out-of-time. As described by Scikit-learn, calibration refers to the relationship between predicted probabilities and event frequencies, and the package offers calibration-curve tools for such assessments.

If resampling is done using SMOTE or any other method, it needs to be done on training data. Since resampling changes the distribution of the classes seen by the model, predicted probabilities cannot automatically be considered as population PDs. Instead, the project needs to use class weighting in comparison with resampling since SMOTE is not always better.

2.5. Expected Loss and decision thresholds

Threshold for classification is a business decision rather than statistical parameter. A threshold set to maximise F1 need not minimise the financial loss since the implications of false positive and false negatives are different from each other.

Expected Loss formula is typically:

$$EL = PD \times LGD \times EAD.$$

The Basel internal-ratings-based approach separates PD, LGD, EAD, and maturity into different risk factors, suggesting keeping PD separate from the others because they form independent components of the Basel internal-ratings-based approach.

For the current project, the connection between PD and the decision rule will be made using the following factors:

- An assumption about the LGD.
- EAD defined as funded or outstanding exposure.
- An interest/income margin assumption.
- Costs of approval and default.
- Reserve/risk-appetite constraint.

Before estimating and validating LGD and EAD, all Expected Loss results have to be stated as illustrative scenarios rather than empirically validated losses.

2.6. Explainable AI

Many machine learning approaches sacrifice interpretability for the sake of flexibility. SHAP is a unified approach to attributing a prediction of a machine learning model to its input features (Lundberg & Lee, 2017). This method allows for:

- Global feature-importance analysis.
- Directional interpretation of the risk contribution.
- Local explanation of individual predictions.
- Comparison of explanations across models.

SHAP explanations cannot be used as proofs of causality. They explain the functioning of the model chosen and trained on a specific set of data and feature representation. SHAP explanations do not provide an automatic satisfaction of the adverse-action requirement, which requires the creditor to state the reasons for such actions clearly and accurately.

Therefore, for the current project, the SHAP methodology should be used as an explanation and diagnostic tool rather than a regulatory-compliant one.

2.7. Fairness and model governance

A fairness analysis is needed since credit decisions may generate disproportionate outcomes within protected groups. Regulation B prohibits discrimination on several aspects of a credit transaction and adverse-action notice must be clear and specific.

The current Lending Club cohort does not provide a proper set of confirmed protected-group variables for a realistic fair-lending audit. Variables such as state, income, employment tenure, or home ownership cannot be loosely renamed as race, sex, or other protected characteristic. In case of synthetic protected groups or their proxies, it needs to be noted that results only demonstrate fairness in an illustrative manner.

Any reasonable extension of fairness assessment requires:

- Institutional protected-group data.
- Ethical and legal approval of its use.
- Group definitions defined in advance.
- Metrics like difference in selection rates, adverse-impact ratio, difference in true positive rate, and difference in false positive rate.

- Intersectional group analysis when possible.
- Discussion of trade-off between performance and fairness constraints.

2.8. Cross-study synthesis and methodological limitations

As seen throughout the literature, there appears to be an ongoing trade-off rather than an algorithm that is better than others. Dastile et al. (2020) conduct a review of statistical and machine learning methods and discuss persistent problems with data preparation and evaluation while Shi et al. (2022) tackle such questions as imbalanced dataset, inconsistencies and lack of model transparency. Suhadolnik et al. (2023) examine ten algorithms on more than 2.5 million institutional data points and confirm high performance of ensembles, especially XGBoost. Such results support the use of nonlinear models but fail to show that ensemble will perform better for other banks, populations and/or evaluation periods.

Another problem is that performance reported in classification metrics is not an indication that probabilities generated by the model would be usable for financial decision-making. As Chawla et al. (2002) showed, SMOTE allows improved learning of an imbalanced minority class. However, the resampled training population is not the same as the population where the bank will apply estimated probabilities. Another relevant point to be made about the problem discussed is that of governance. Alonso and Carbó (2020) argue that while predictive power of machine learning could bring significant benefits, it should be evaluated considering additional resources needed to manage increased complexity of such models. Thus, this project assumes that class balancing, resampling for training only, probability calibration and out-of-time evaluation are not automatically solved by improvement in classification metric.

In addition, research differs in the way they describe the decision process to be used. The model that is being trained on originated loans cannot prove performance for rejected applications when actual repayment results are unknown. Moreover, explanation of the predicted probability cannot prove causation or fairness to protected groups. These limitations shape the scope of the present project: its principal empirical claim will concern observed outcomes among selected Lending Club loans, not all applicants or regulatory compliance.

2.9. Current research frontier

Recent literature considers several criteria when assessing modern credit scoring techniques. Aboulmaouda et al. (2026) construct their systematic study according to the criteria of accuracy,

stability, and transparency of model, keeping Logistic Regression as a representative example of the last category. Montevechi et al. (2024) argue that the improved process of modelling and interpretability are necessary preconditions for the successful application of advanced classifiers. Thus, not only development of new classifiers is an essential part of research frontier, but also establishing their reliability and ability to explain their predictions.

Explainability plays the key role in distinguishing both concepts. Hadji Misheva et al. (2021) use LIME and SHAP to provide explanations for Lending Club models and consider issues of implementation of the approach. Pathi and Pothineni (2025) compare SHAP, LIME, and hybrid approaches to explainability, focusing on consistency and repeatability of explanation procedure. Hardt et al. (2016), answering the related problem, show that the criteria of group-fairness need information on target, prediction, and membership to the group. Overall, the above-mentioned works illustrate that the quality of explanation, fairness testing, and accuracy of prediction are different criteria; not one of them replaces another.

The current project presents some fundamentals concerning data, target, feature-timing, and temporal validation of the models. Model calibration and explanation procedures are the elements of modelling phase. Fairness audit of protected groups and LGD/EAD system are beyond the scope of claims currently supported by this public cohort.

2.10. Project positioning

Further model comparison based solely on the analysis of Lending Club's dataset is unlikely to bring a unique contribution. For example, Chen's comparison study for the family of models on credit card data in Taiwan considers many other papers that extensively compare various statistical models, machine learning, and deep learning methods (Chen, 2024; Dastile et al., 2020; Aboulmaouda et al., 2026). However, the distinguishing feature of the proposed study is its research design: a precisely defined resolved loan outcome, chronological training and testing phases, separation of repayment from predictors, and comparison between borrower/credit history characteristics and Lending Club's own grade and price outcomes.

The previous OpenIntro dataset is relevant to start the project with, but not the source for the current modelling results. Specifically, OpenIntro classifies loans_full_schema as a dataset with 10,000 resolved loans that were originated via Lending Club; in contrast, in the current study, the data of 603,589 loans are collected based on George's larger Kaggle dataset (OpenIntro, n.d.; George, 2019). Thus, the latter allows for a more detailed analysis of history but does not exclude selection bias

because both datasets involve the data about originated loans and not the results of the rejected applications.

In the end, it is necessary to prove the contribution not through the improved pipeline but through findings. The model evaluation stage is supposed to estimate whether a nonlinear model performs better than a clear baseline; whether it keeps improving in terms of discrimination and calibration on the following year; and the difference in performance if the grade and pricing predictors provided by lenders are included. These are empirical questions, not results already established by the current preparation notebooks.

2.11. Research gap and project contribution

The previous work shows the standard credit classification approach on a small snapshot using an immature target and less realistic validation scheme. This updated work fills in the narrower, but more justified methodology gap:

> Creation and evaluation of a reproducible historical credit-risk pipeline with focus on outcome maturity, temporal validation, leak prevention, probability calibration, and comparison of borrower-level features to lender-generated pricing features.

Therefore, it is not about construction of a production loaning system in any way. It is just a methodological example of how realistic and auditable credit risk machine learning research can be done before modeling.

3. Competitive analysis

3.1. Business and technical context

The credit risk framework under analysis faces the competition of several major approaches:

1. Standard scorecard approaches and Logistic Regression.
2. Benchmarks based on academic machine learning, including non-linear models.
3. Automated machine learning and AI commercial applications.
4. Bank or fintech internal model development tools.
5. The proposed approach to the reproducible research with explanations.

The comparison must be made based on predictability, explainability, calibration, governance, cost, and reproducibility rather than on the suggestion that a student project could replace a regulated lending application.

3.2. Business need and intended users

Target users of the framework include credit-risk analysts creating challenger models, model-validation teams evaluating assumptions and data, and portfolio decision-makers distinguishing performance from reliability of predicted probability. If a bank is considering switching from a transparent scorecard to a more versatile ensemble model, decision-making cannot stop at whether the new model performs better but needs to account for cost-effective validation and governance as well. Alonso and Carbó (2020) describe this as the trade-off between prediction and supervisory cost.

The practical utility of the proposed framework consists in providing an audit trail from a documentable historic cohort to the out-of-time model comparison. An analyst may find out which loans qualify, what variables have been removed from the PD prediction, what transformations have been learned from training data, and performance dynamics in a following issue-year cohort. The scope is smaller than the commercial lending platform as no live application flow has been added, no outcomes for the rejected applicants provided, and no institutional policy enforcement or protected group monitoring has been included.

An understandable rationale exists for undertaking documented challenger analysis of this kind; however, there is no proof of market viability through customer interviews, cost analysis, and pilot program deployment. Hence, the competitive argument ought to be made on the basis of methodology: a transparent process that enables the testing of assumptions about validity is preferable to a model comparison which cannot be explained. It must not be made as an alternative to a proper lending system.

3.3. Comparative analysis

Approach	Strengths	Weaknesses	Relevance to this project
Logistic Regression / scorecard	Transparent coefficients; easy benchmarking; familiar governance; relatively simple calibration	May miss nonlinear interactions; requires careful transformations and feature design	Primary interpretable benchmark
Random Forest / gradient boosting	Captures nonlinear effects and interactions; often strong tabular performance	More difficult to explain directly; can require calibration and tuning; may be unstable under drift	Candidate nonlinear comparison
Naive academic ML benchmark	Easy to implement; low software cost; useful for teaching	Often uses random splits, leakage-prone preprocessing, immature labels, and accuracy-focused metrics	The revised workflow is designed to correct these weaknesses
Commercial AI platforms	Integrated deployment, monitoring, governance, and experimentation tools	Licensing cost, vendor dependence, restricted transparency, and implementation complexity	Industry comparator, not a direct numerical benchmark
Internal institutional scorecard	Access to proprietary data, policies, protected-group monitoring, and operational outcomes	High development, governance, and data-integration requirements	More realistic production reference
Proposed framework	Reproducible open-source workflow; historical cohort selection; chronological validation; separate outcome table; explainability-ready architecture	Not production validated; no live data; no verified protected-group data; LGD/EAD not yet empirically modelled	Academic proof of concept and methodological framework

3.4. SWOT analysis

Framework	SWOT Category	Details
Traditional scorecard	**Strengths**	High transparency. Easy to document and challenge. Familiar to credit-risk practitioners. Suitable as a benchmark.
Traditional scorecard	**Weaknesses**	May underrepresent nonlinear interactions. Requires careful binning, transformations, and stability monitoring. Can be less flexible for complex feature relationships.
Traditional scorecard	**Opportunities**	Combine with monotonic constraints, scorecard development, and calibration. Use as a challenger model against boosting.
Traditional scorecard	**Threats**	May be outperformed on discrimination by well-governed ensemble models. Can become outdated when borrower populations or economic conditions change.
Academic machine-learning benchmark	**Strengths**	Rapid experimentation. Broad algorithm comparison. Low licensing cost. Useful for methodological teaching.
Academic machine-learning benchmark	**Weaknesses**	Random splits can overstate future performance. Preprocessing can leak information. Accuracy or F1 may be prioritised over PD calibration. Static thresholds may ignore financial costs. Explanations may be added after modelling without connecting them to governance.
Academic machine-learning benchmark	**Opportunities**	Improve using out-of-time validation, calibration, leakage controls, and decision-cost analysis.
Academic machine-learning benchmark	**Threats**	Misleading reported performance. Poor reproducibility. Inappropriate deployment or regulatory interpretation.
Commercial AI platform	**Strengths**	Integrated experiment tracking, deployment, monitoring, and governance. Potentially faster operationalisation. Support for model comparison and workflow automation.
Commercial AI platform	**Weaknesses**	Cost and vendor lock-in. Proprietary components may limit independent auditability.

		Outputs still depend on the institution's data, target, and validation design.
Commercial AI platform	**Opportunities**	Used as an implementation environment after a sound research methodology has been established.
Commercial AI platform	**Threats**	Automation can create false confidence. Governance weaknesses may be hidden behind a polished interface.
Proposed framework	**Strengths**	Uses 603,589 resolved 36-month loans rather than the earlier small snapshot. Explicitly defines the target. Uses chronological train, validation, and test periods. Separates PD inputs from repayment outcomes. Stores raw, processed, feature-rule, and preprocessing artifacts. Compares a borrower/application baseline with a lender-pricing feature set. Supports calibration, SHAP, and later Expected Loss analysis.
Proposed framework	**Weaknesses**	Historical Lending Club data do not represent all lenders or all applicants. The cohort contains originated loans only. The target is final observed status, not a fixed-horizon PD. No full protected-group fairness audit is currently possible. LGD and EAD are not yet empirically validated. The current system is a reproducible research pipeline, not a production lending platform.
Proposed framework	**Opportunities**	Add calibrated PD models. Compare Logistic Regression with LightGBM or another ensemble. Add SHAP explanations. Estimate or scenario-test LGD and EAD. Add population-stability and performance-drift monitoring. Add fairness analysis when appropriate governed data are available.
Proposed framework	**Threats**	Label and cohort selection bias. Vintage and economic drift. Data-source changes. Misinterpretation of illustrative fairness or Expected Loss assumptions. Overclaiming regulatory compliance from explainability alone.

The current project aims at being a reproducible credit risk modeling framework for academic purposes rather than a production lending application. The most important innovation of the current project is its methodology, which uses the advanced approach in which the initial approximately 10,000 loan dataset is substituted by the selected 603,589 loans dataset of Lending Club loans that have been resolved within 36 months; introduces a clear observed outcome target variable; separates the repayment outcomes from the probability of default predictors; uses out-of-time validation; and preserves the preprocessing artifacts in training only mode.

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